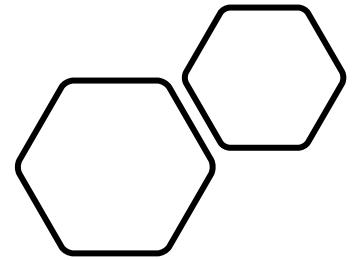


Head-to-Head Performance of Commercially Available Deep Learning Systems for Diabetic Retinopathy Screening: A Systematic Review and Bayesian Network Meta-analysis



Presenter: Ting-Wei, Wang MD PhD

Advisor: Yu-Bai Chou MD PhD

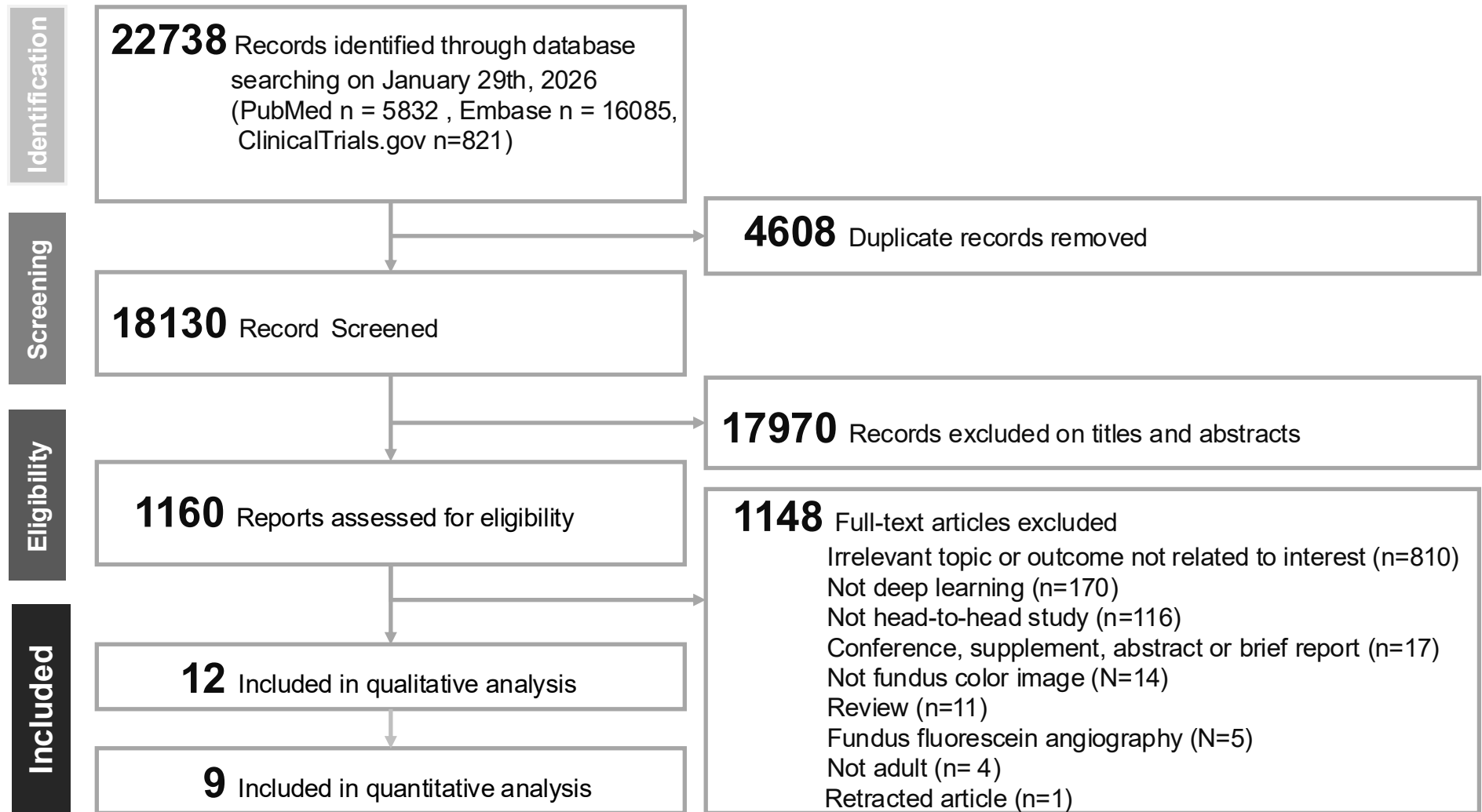
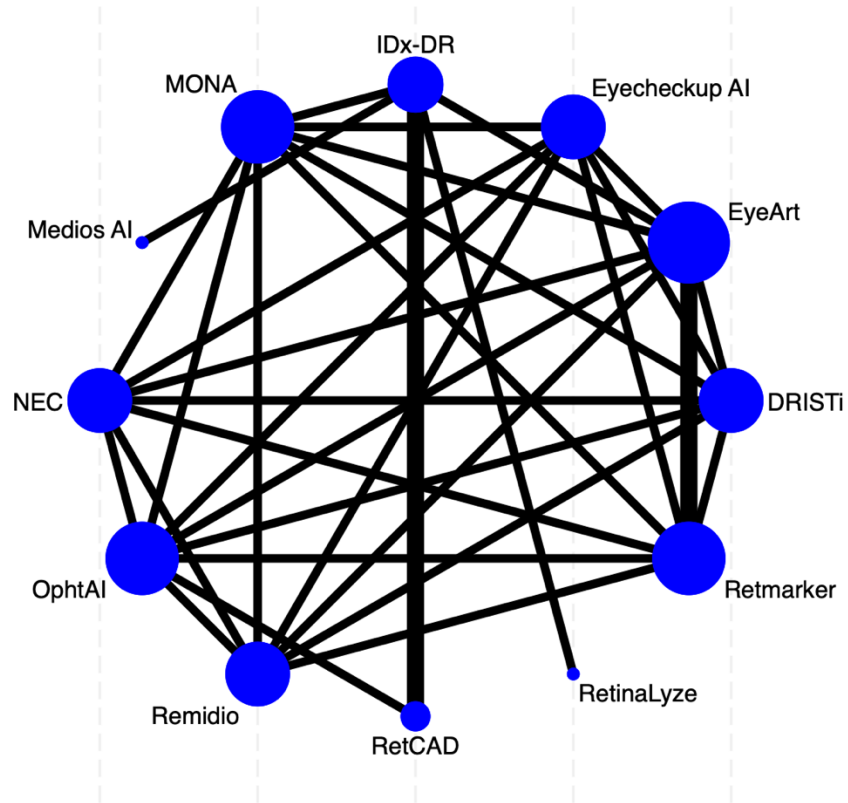


Figure 1. PRISMA flowchart of included studies

First Author (Year)	Country	Study Design	Sample Size (Patients)	Age (Mean/Median)	Target Condition	Reference Standard	Camera Type / Brand	AI Algorithms Evaluated
Bauer (2025)	Germany	Prospective	143	56 yrs	Referable DR (Mod+)	Ophthalmologist (ICDR)	Handheld (Optomed Aurora)	RetCAD, OphtAI
Huhtinen (2025)	Finland	Prospective	156	NR	Referable DR (Mod+)	2 Ophthalmologists	Handheld (Optomed Aurora)	10 Anonymized (Top 10 of 25)
Poschkamp (2025)	Germany	Observational	1,716	50.4 yrs	Referable DR (>Mild)	Mydriatic Funduscopy (6 Ophth)	Desktop (Topcon TRC-NW400)	IDx-DR, RetCAD
Rudnicka (2025)	UK	Observational	126,365	60.5 yrs	Referable DR (M1/R2/R3)	NHS Protocol (up to 3 graders)	Desktop (Approved range)	8 Systems (EyeArt, OphtAI, etc.)
Wroblewski (2025)	Mexico	Retrospective	248	~56 yrs	DR	2 Retinologists	Smartphone (Remidio FOP)	EyeArt, Medios
Grzybowski (2025)	Poland	Retrospective	758	NR	Referable DR	3 Retina Specialists	Desktop (Topcon TRC-400NW)	IDx-DR, RetCAD
Grzybowski (2024)	Poland	Retrospective	695	57 yrs	DR / Referable DR	3 Retina Specialists	Desktop (Topcon TRC-400NW)	IDx-DR, Mona-DR
Musetti (2024)	Italy	Retrospective	201	65 yrs	Referable DR	2 Retina Specialists + OCT	Desktop (Topcon DRI 3D OCT)	IDx-DR, EyeArt
Grzybowski (2023)	Poland	Retrospective	811	NR	Referable DR	3 Retina Specialists	Desktop (Topcon TRC-400NW)	IDx-DR, Medios
Grzybowski (2021)	Poland	Retrospective	170	≥18 yrs	Referable DR	Single Ophthalmologist	Desktop (Topcon TRC-400NW)	IDx-DR, Retalyze
Lee (2021)	USA	Retrospective	23,724	62.7 yrs	Referable DR (Any DR)	Arbitrated Panel (3 Specialists)	Desktop (Topcon TRC-NW8)	7 Masked (OphtAI, EyeArt, etc.)
Tufail (2017)	UK	Prospective	20,258	60 yrs	Referable DR	NHS Graders + Arbitration	Desktop (Mixed)	EyeArt, Retmarker

Network geometry

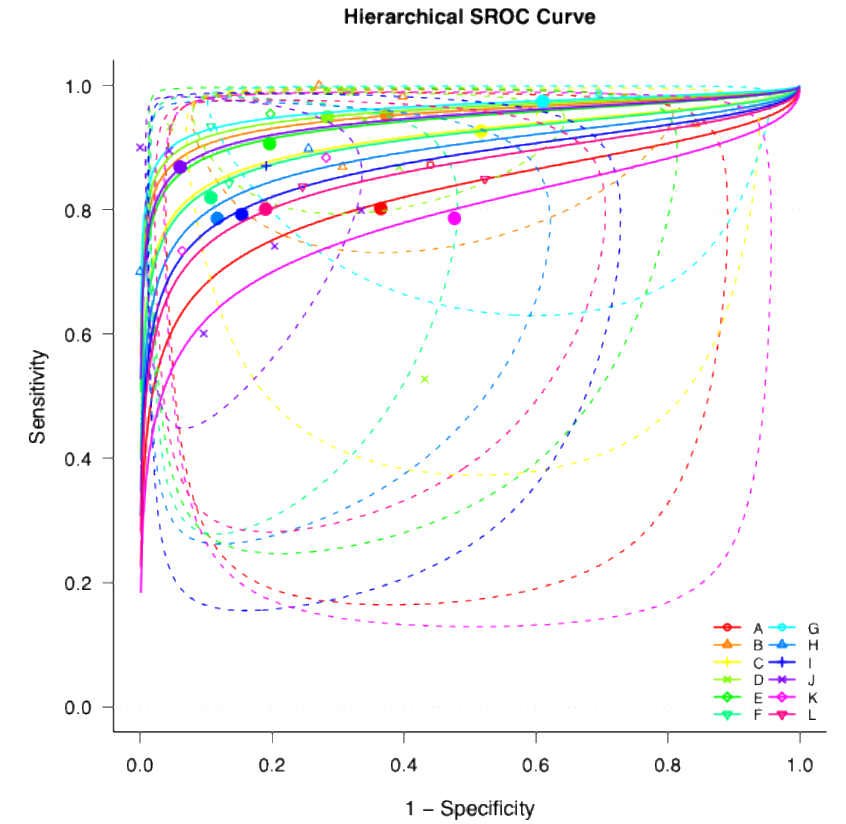
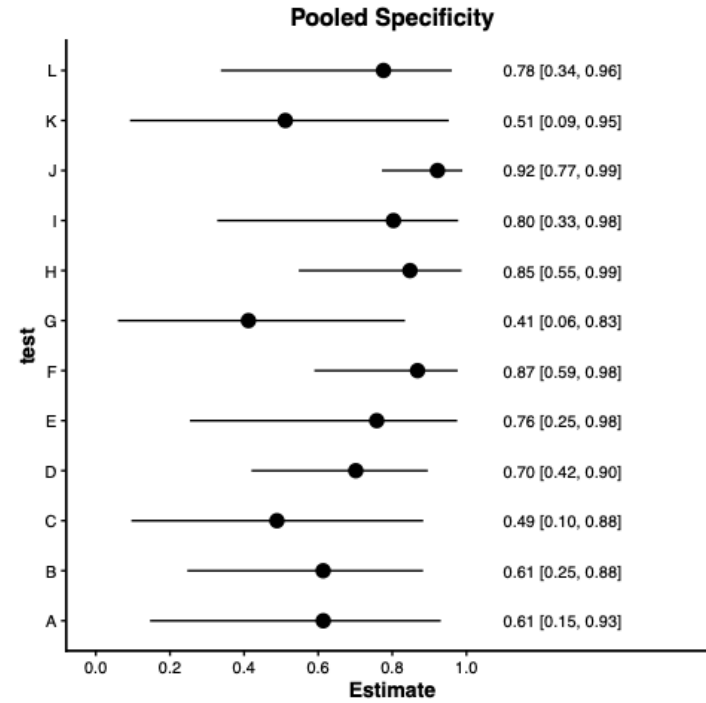
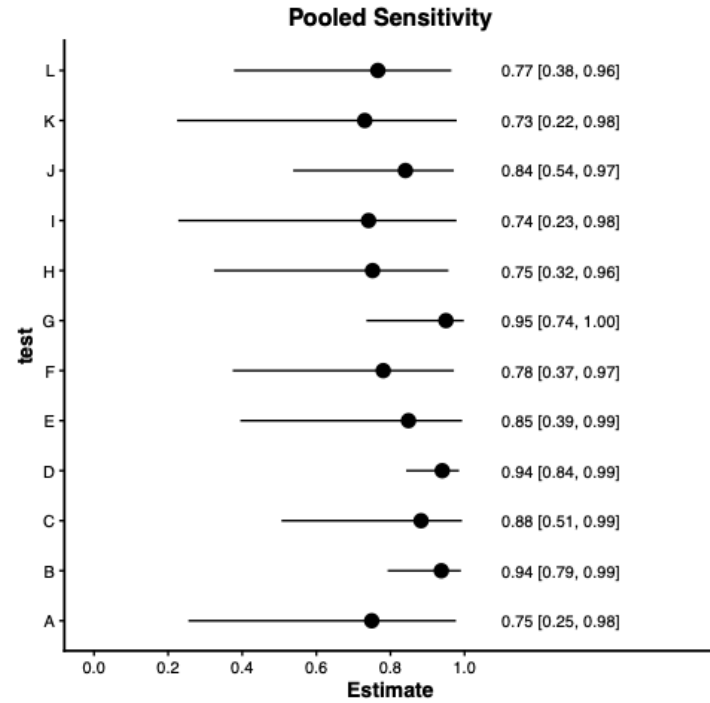


Inconsistency test:

Global design-by-arm test: 0.9472

Explore loop inconsistency: 0.9472

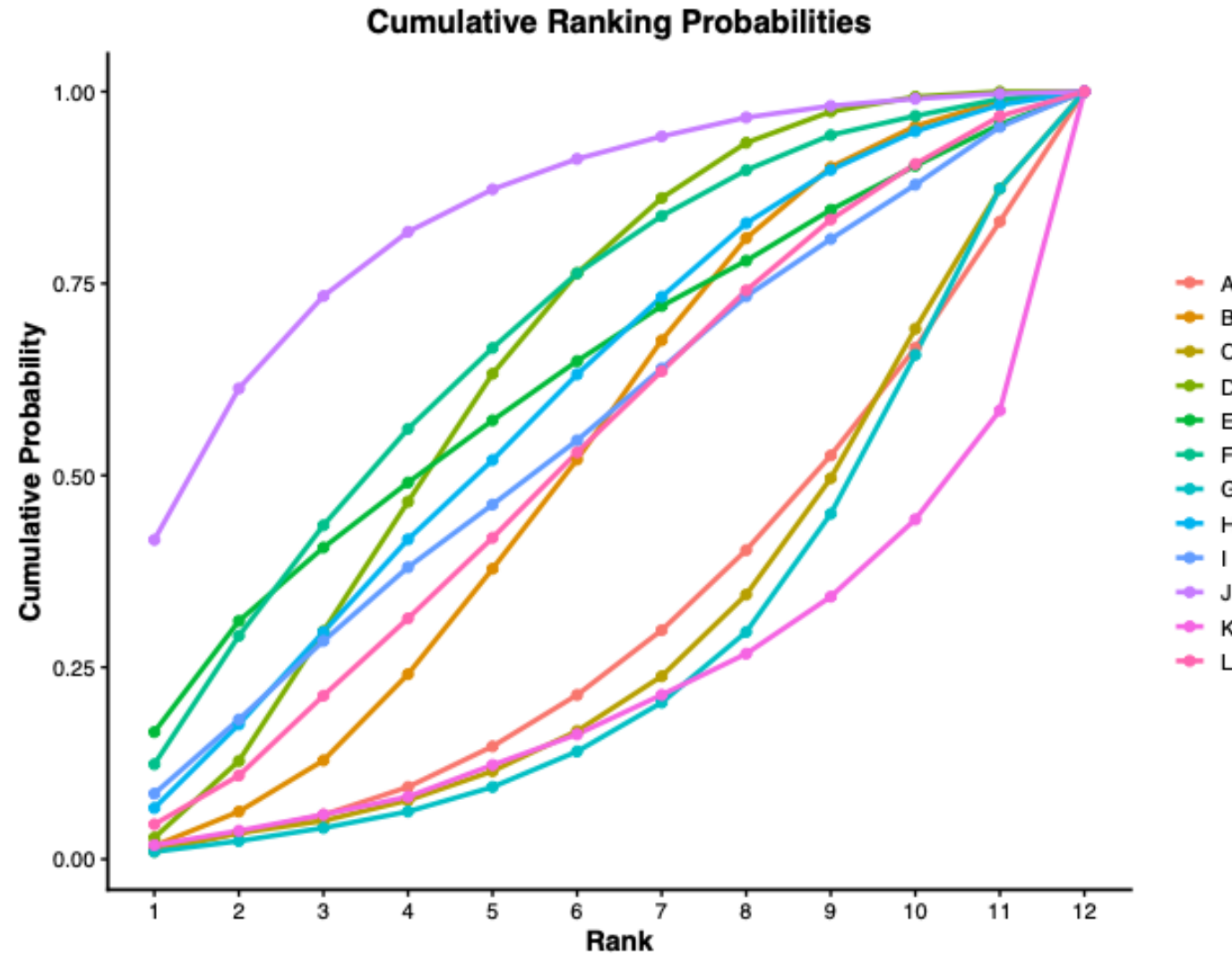
Overall



Subgroup Analysis

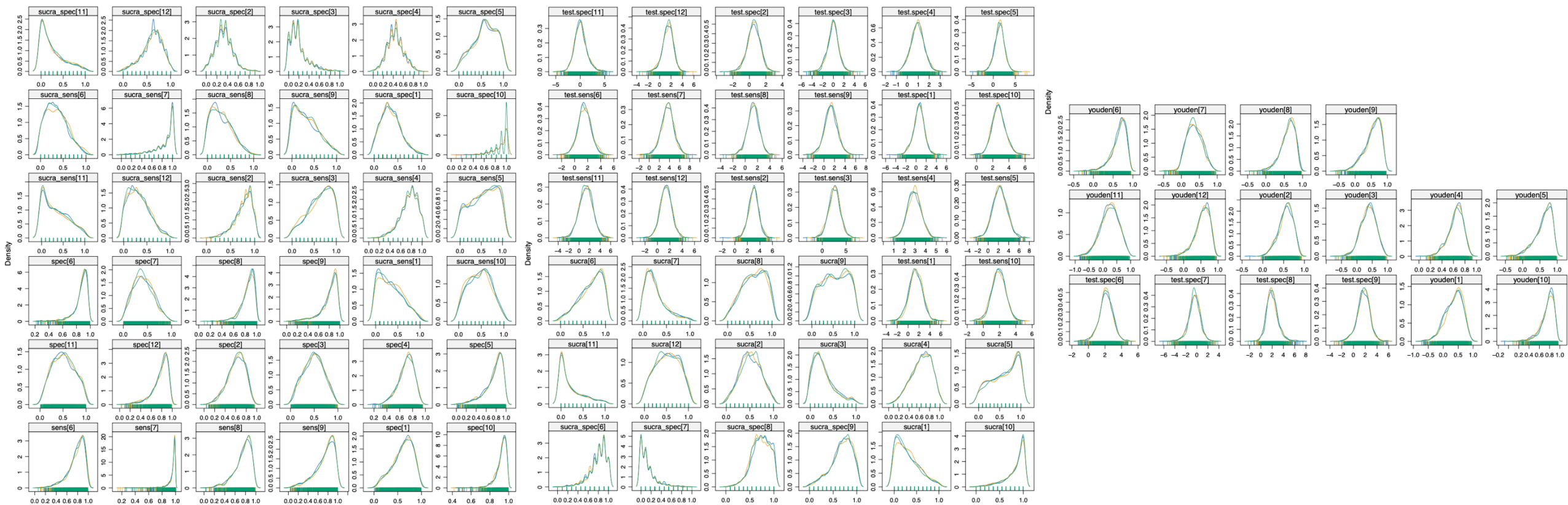
Covariate	Sens_Beta	Sens_LCI	Sens_UCI	Sens_Sig	Spec_Beta	Spec_LCI	Spec_UCI	Spec_Sig
prevalence	-0.0273839071954748	-0.314117744250342	0.25858804393085	NS	0.0719767682626038	-0.199525884606201	0.322591814908104	NS
economical	1.67187993149295	-3.70377704965394	7.33996788330351	NS	0.780999470357062	-4.89790148760542	6.22213838377872	NS
healthcare	-0.308394367187514	-0.826543026145927	0.228501820357743	NS	-0.0283836306872263	-0.546076607191664	0.464075624014066	NS
cameratype	-0.0838794816700779	-0.546820606660842	0.37973848574192	NS	-0.232946842626289	-0.689926403381401	0.252771410858152	NS
field	-0.227884468691449	-0.484163179486826	0.0552921036058175	NS	0.15098267356293	-0.115147552486882	0.435834186408344	NS
vendor	-0.238727326575967	-0.89287285745834	0.419174626050919	NS	-0.0109470926027146	-0.601019518279905	0.626141830630373	NS
gradable	0.0771417245680998	-0.0811753211990645	0.221184427987206	NS	-0.0948818953535488	-0.258549719265634	0.0634863866665969	NS
dilation	-0.0495089758698866	-0.518214441662716	0.389657108715175	NS	-0.0672399694696509	-0.503979983992696	0.369058065052779	NS
diagnostic criteria	0.0997574745575002	-0.17619793145771	0.372662747709901	NS	-0.119724998271147	-0.369354782964207	0.149058807180639	NS

SUCRA Plot



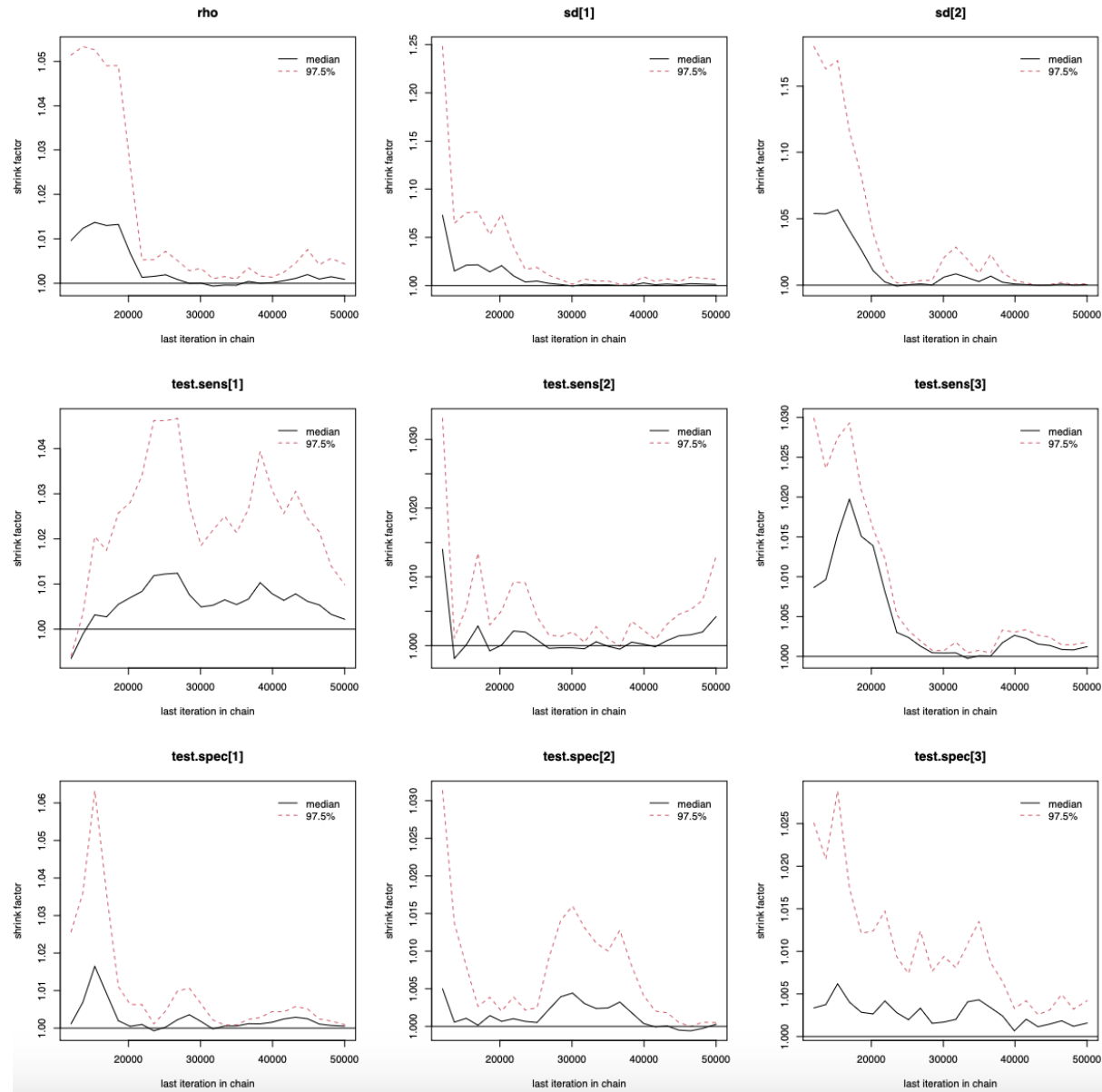
A	DRISTi
B	EyeArt
C	Eyecheckup AI
D	IDx-DR
E	Medios AI
F	MONA
G	NEC
H	OphtAI
I	Remidio
J	RetCAD
K	RetinaLyze
L	Retmarker

Posterior probability



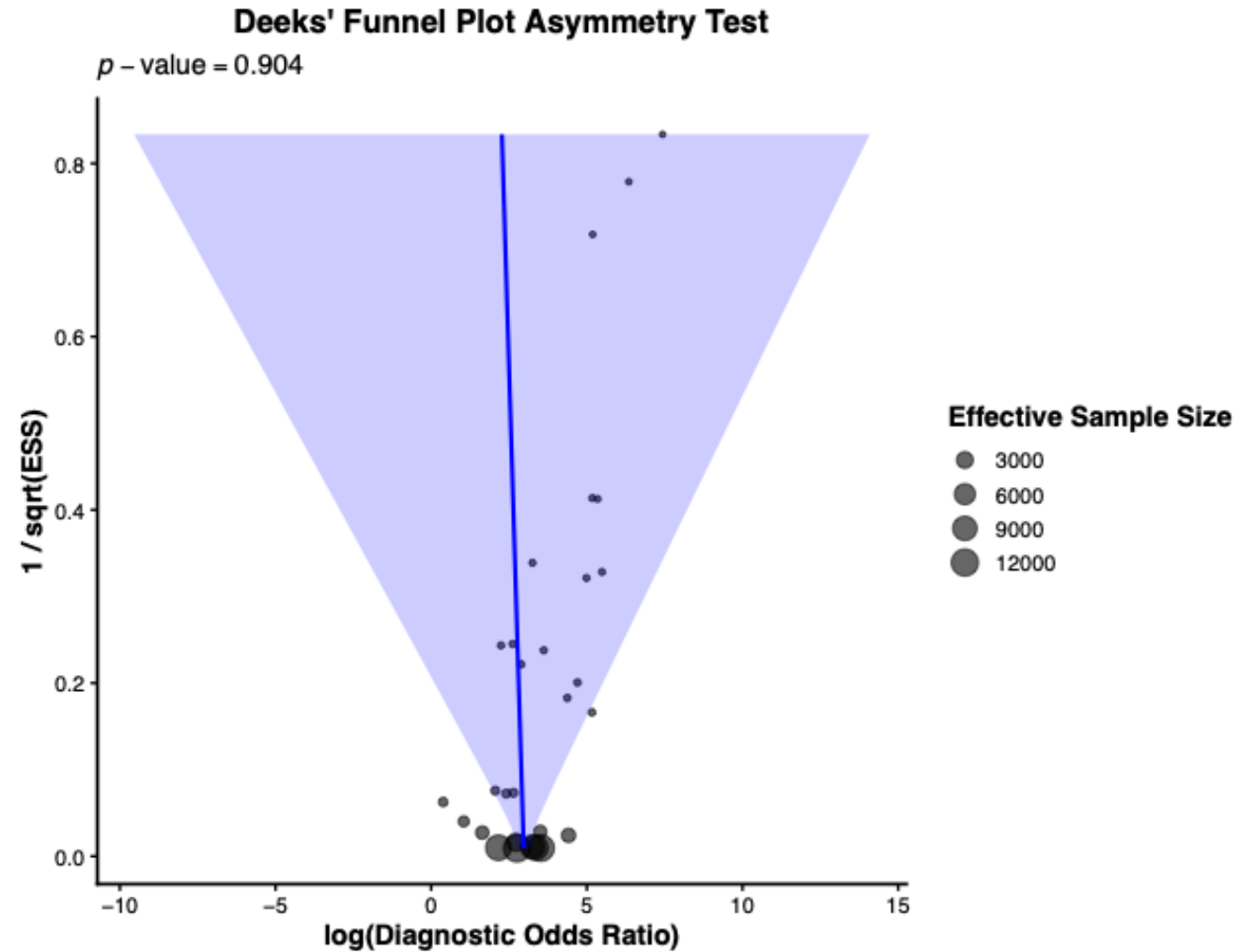
No class switching

Convergence



Converge: <1.1

Publication bias: Deek's funnel plot



Discussion

- Bayesian network meta-analysis method for complex model estimation with relative limited data
- (+) First Bayesian network meta-analysis of head-to-head comparison commercial model (Originality)
- (-) Limit studies available currently (Limit evidence)
- Target: Ophthalmology? JAMA Ophthalmology? American journal of Ophthalmology?